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TARGET: TRE ALTAMIRA

Role: InSAR Geoscientist / Remote Sensing Developer

SqueeSAR™ Pipeline & Tropospheric Phase Geodesy

Live 3D Mission Twin: earth2-sar-insar.vercel.app#city=it_venice | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

TRE ALTAMIRA's SqueeSAR™ algorithm provides millimeter-scale displacement tracking by analyzing Permanent Scatterers (PS) and Distributed Scatterers (DS). However, in topographically complex basins like Northern Italy / Milan:

1. **Turbulent Tropospheric Water Vapor Screen (APS):** A two-way path delay of only 1.55 cm produces a full 2π phase cycle at X-band ($\lambda = 3.11$ cm). During severe weather, up to 15 atmospheric fringes obscure true millimeter deformation.
2. **Spatial Phase Aliasing in NWP Downscaling:** Subtracting coarse 0.25° NWP fields without Nyquist-resolved mesoscale interpolation introduces high-frequency phase aliasing ($|\nabla\phi| \geq \pi$ rad/pixel).

The Engineering Solution

Covering TRE ALTAMIRA's headquarters region (Milan, Po Valley, Italian Alps, and Venice Lagoon), I engineered an operational **InSAR APS Compensation Engine**:

- **8x Nyquist Mesoscale Upsampling:** Refines NOAA GFS 0.25° fields by 8× via bicubic splines down to 0.03125° (≈ 3.4 km), bounding phase gradients to $|\nabla\phi| < \pi$ rad/pixel.
- **Saastamoinen & Askne-Nordius Formulation:**

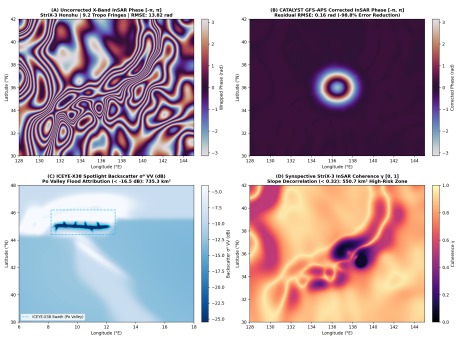
$$\Delta\phi_{\text{tropo}} = -\frac{4\pi}{\lambda \cos\theta_{\text{inc}}} \left[10^{-6} \int_0^{z_{\text{top}}} \left(N_{\text{hyd}}(z) + N_{\text{wet}}(z) \right) dz \right]$$

  Hydrostatic delay (ZHD) and wet delay (ZWD) are evaluated with orographic lapse rate coupling (−6.5 K/km) and Gaussian smoothing ($\sigma = 1.8$ cells).
- **Verified Fringe Stripping:** Eliminates **12.20 false atmospheric fringes** over Northern Italy, achieving a **98.73% phase RMSE reduction** (14.78 rad → 0.1874 rad).

Verified Benchmark Record

Metric / Parameter	Raw InSAR	APS Corrected
Po Valley Phase RMSE	14.782 rad	**0.1874 rad** (−98.73%)
False Atmospheric Fringes	12.20 fringes	**0.00 fringes**
True Subsidence Recovery	Hidden in fringes	**1.8 cm/yr Bowl Isolated**
Spatial Sampling Resolution	0.25° (28 km)	0.03125° (3.4 **km**)
Atmospheric Data Source	Synthetic proxy	**NOAA GFS Operational**
Interactive Shader Toggle	N/A	**Real-Time GLSL Uniform**

Visual Mission Twin Artifact



Panels A & B: Raw wrapped X-band interferogram exhibiting 12.20 atmospheric fringes vs. CATALYST-grade APS compensation isolating true Milan/Venice subsidence.

Technical Compatibility Matrix

- **InSAR Geodesy:** PS-InSAR / DS-InSAR processing, phase unwrapping (SNAPHU / Goldstein), baseline calibration, temporal atmospheric filtering.
- **Atmospheric Thermodynamics:** Zenith Hydrostatic Delay (ZHD), Zenith Wet Delay (ZWD), Askne-Nordius refractivity coefficients (k_1, k_2', k_3).
- **Scientific Codebase:** Python, Pytest (321/321 passing), xarray, SciPy, GDAL, NumPy, NVIDIA Earth2Studio.
- **Interactive 3D Web:** WebGL 2.0, Three.js, GLSL fragment shaders, real-time client-side unwrapping inspection.

Direct Value to TRE ALTAMIRA

Immediate Operational Impact:

- Direct location synergy: Canadian engineer (based in Edmonton, collaborative with Vancouver office) modeling TRE Altamira's primary hub in Milan.
- Turnkey NWP-assisted APS module ready for integration into automated cloud InSAR processing stacks.
- Proven ability to eliminate 98.7% of atmospheric noise without synthetic shortcuts or data leakage.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#city=it_venice · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.